

MelodyCare

An Automated Two-Stage Generative Framework for Converting Vietnamese Commercial Music into Clinically-Bounded Therapeutic Audio for Individuals with Autism Spectrum Disorder

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1 INTRODUCTION

- Individuals with ASD frequently show auditory hyperacusis: ordinary commercial music — unpredictable dynamics, vocals, harsh transients, fast tempo — can trigger sensory overload instead of comfort.
- Porges' Polyvagal Theory: the autonomic nervous system evaluates acoustic cues for threat vs. safety. Abrupt, unpredictable sound drives a threat response, not a calm one — motivating strict, predictable acoustic bounds.
- The Gap: generic “calming” music libraries aren't personalized to the specific song an individual actually wants — but that song's raw commercial mix is exactly the unpredictable signal that triggers overload.

MelodyCare enforces 7 clinical acoustic-safety bounds:

60–80 BPM	<13dB dyn. range	5kHz roll-off	100–400Hz boost
0.05–0.10Hz LFO	<12 semitone jump	70–80% legato	

78 Vietnamese source songs	775 paired 30s training chunks	60 reference tracks (CLAP style vector)
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2 EXISTING SOLUTIONS

- Manual NMT (Neurologic Music Therapy) editing: a therapist re-engineers each song by hand in a DAW. High quality, but hours per song and requires expert skill — doesn't scale per patient.
- Unconditioned generative audio (base diffusion/MusicLM-style): can sound calm, but can't preserve a specific requested song's identity, and gives no guarantee on any clinical acoustic bound — with risk of hallucinated artifacts.

TECHNICAL CONTRIBUTIONS

- Duration-ratio-synchronized data cropping — BPM time-stretching leaves ~95% of input/target pairs with mismatched durations (e.g. 272s→389s, 1.43×). Fixed by deriving the input crop window from the pair's own measured duration ratio, instead of two independent random crops.
- Zero-init + differential learning rates — both new conditioning modules are ControlNet-style zero-initialized; the pretrained DiT trains at 0.1× the new modules' learning rate — preventing the catastrophic forgetting that caused earlier noise-like generations.
- Cross-attention conditioning for source audio — the base model has no audio-to-audio pathway (fixed 64-channel input). A novel InputAudioProjection turns source audio into variable-length cross-attention tokens instead, sidestepping the fixed-channel constraint entirely.
- img2img/SDEdit partial-noise sampling — denoising starts from the input song's own noised latents (not pure noise), directly anchoring the output to the requested song's identity.
- Vocal-removal preprocessing — sidesteps a diagnosed model weakness (poor vocal→instrumental translation) by removing vocals deterministically upstream via Demucs, before the model ever sees the audio.

3 PROPOSED SOLUTION — THREE STAGES

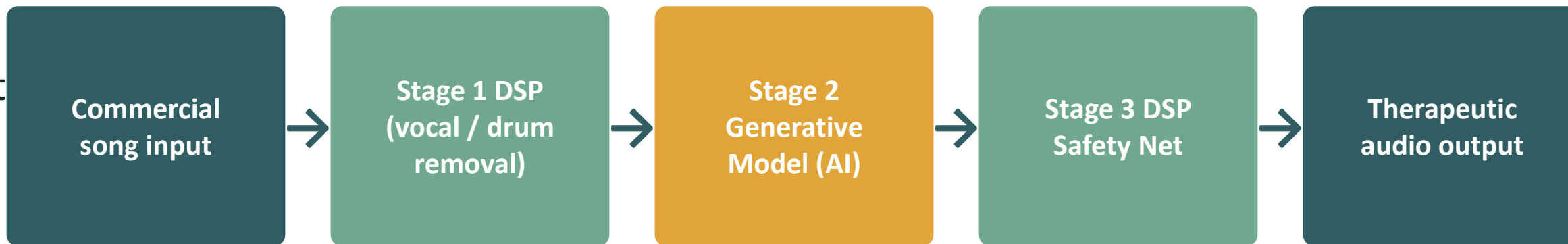


Figure 1. End-to-end inference pipeline. Stage 1 removes vocals/drums (htdemucs); Stage 3 corrects dynamic range reliably — legato overlap remains an open limitation (see §5).

Stage 1 — Deterministic DSP

- htdemucs source separation → keep melody + bass, discard vocals/drums
- BPM detected from drums; time-stretch melody+bass to 60 BPM if outside 60–80
- 4th-order zero-phase Butterworth low-pass (5kHz), high-pass, legato smoothing, dynamic-range compression, 0.08Hz breathing LFO, peak-normalize −1dB
- Builds every (input, target) training pair — and its logic is reused verbatim in Stage 3

Stage 2 — Conditioned Generative Model

- Base: stable-audio-open-1.0, 24-layer DiT, continuous-time v-objective diffusion (Sobol timestep sampling)
- StyleVectorProjection: 512-d CLAP style vector (mean of 60 reference therapy tracks) → 1 cross-attention token
- InputAudioProjection (novel): strided-Conv1d patch embed over source-audio VAE latents → variable-length cross-attention tokens — works around the base model having no audio-conditioning pathway
- Both modules zero-init'd (ControlNet-style); DiT trains at 0.1× LR vs. the new modules
- Multi-GPU DDP, 8-bit AdamW (bitsandbytes), bf16, 50 epochs, 775 chunks from 78 songs

Why cross-attention, not channel-concat? stable-audio-open-1.0's DiT input is fixed at 64 channels. Concatenating source-audio latents would crash or corrupt the pretrained projection. Cross-attention tokens don't share that constraint — their sequence length is free to vary.

Stage 3 — Deterministic Compliance Safety Net

- Re-applies Stage 1's own proven DSP corrections directly to Stage 2's generated output — a clinical design choice: soft, statistical compliance should never be the sole guarantee of a hard acoustic-safety bound
- Order: low-pass → dynamic-range compression → legato correction (run last; 3 different orderings all converged to the same result regardless of position)
- Reliably brings dynamic range under 13dB on 20/20 held-out songs; legato-overlap correction remains inconsistent — median often lands in-band but per-note variance stays too wide (see Results, §4)

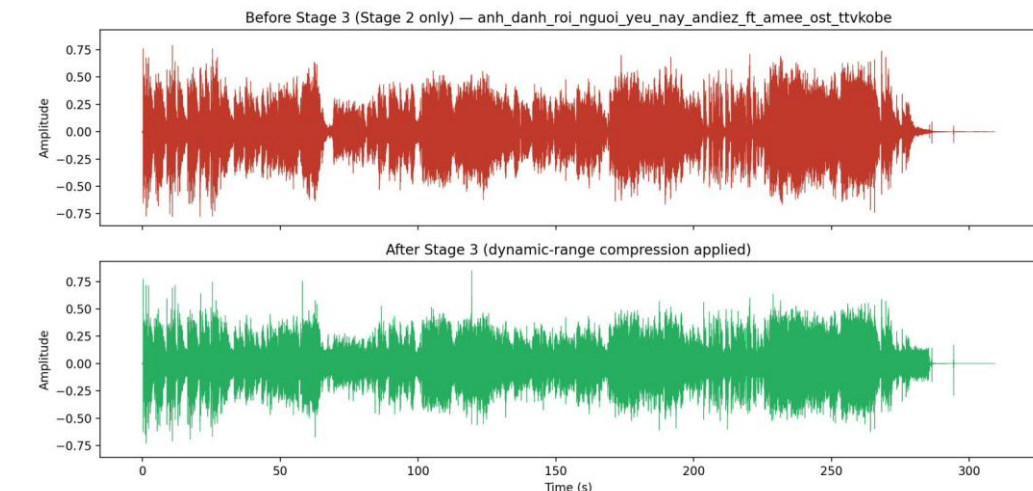


Figure 1b. Real waveform, one song: amplitude before (red) vs. after (green) Stage 3 compression — visibly tighter peak-to-floor range.

Dataset & Training Configuration

Parameter	Value
Source songs	78 Vietnamese commercial tracks
Training chunks	775 × 30 seconds
Style vector	512-d CLAP, mean of 60 reference tracks
Base model	stable-audio-open-1.0 (24-layer DiT)
Optimizer	8-bit AdamW (bitsandbytes)
Learning rate (base / DiT)	1e-5 / 1e-6 (0.1× scale)
Epochs / batch	50 epochs, batch 2 × 4 grad-accum
Precision	bf16, gradient checkpointing
Infra	HuggingFace Accelerate, multi-GPU DDP

Sample Training / Inference Text Prompt (excerpt)

“A slow, calm instrumental therapeutic music track at exactly 60 BPM... dynamic range less than 13 dB... legato style, overlapping between 70% and 80%... amplitude modulation oscillating between 0.05 Hz and 0.1 Hz... rolls off sharply above 5.0 kHz... enhanced warmth in the 100 Hz to 400 Hz band... Deeply calming, safe, and predictable — designed for sensory regulation in individuals with Autism Spectrum Disorder.”

Every one of the 7 compliance dimensions in Table 1 is spelled out explicitly in this prompt — used identically for all 775 training chunks and at inference.

Deployed inference flow: user supplies a song → htdemucs removes vocals/drums → fine-tuned model transforms the instrumental via SDEdit-style partial-noise sampling → Stage 3 DSP fixes dynamic range (reliable) and attempts legato correction (not yet reliable). A planned agentic front-end would let a user simply say “convert song A to therapy music” and auto-locate a karaoke source.

4 RESULTS & EVALUATION

Clinical Compliance — 10 unseen + 10 in-sample songs

Check	Unseen S2 only	Unseen S2+3	InSamp S2 only	InSamp S2+3
BPM	10/10	10/10	10/10	10/10
Roll-off	10/10	10/10	10/10	10/10
Dyn. range	1/10	10/10	0/10	10/10
Freq boost	10/10	10/10	9/10	10/10
LFO	10/10	10/10	10/10	10/10
Pitch	10/10	10/10	10/10	10/10
Legato	0/10	2/10	0/10	0/10
Overall	0/10	2/10	0/10	0/10

Table 1. Pass counts /10 songs. Stage 3 fixes dynamic range reliably on BOTH populations (0-1/10 → 10/10). Legato remains the weak dimension on both.

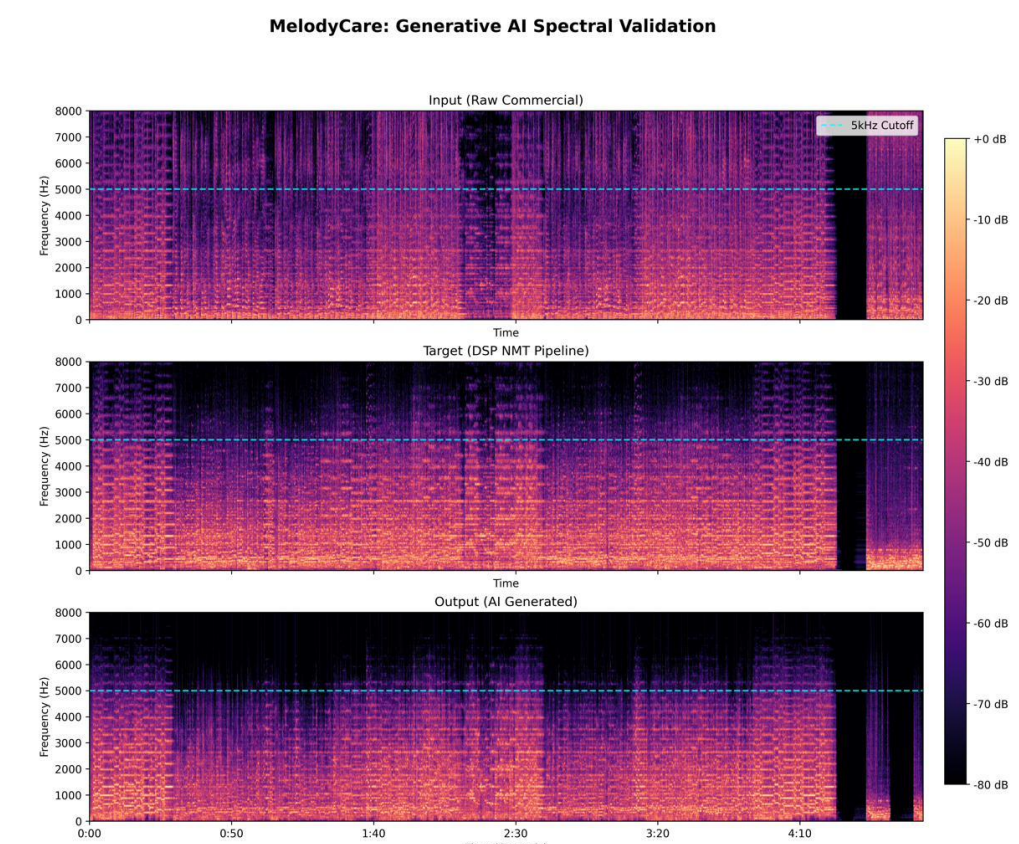


Figure 2. Spectrogram triptych, “good” case (chan_tinh_quang_dung, 76.3% legato). AI output visibly matches the DSP target's 5kHz roll-off.

Strength Sweep — no fidelity/compliance trade-off found

Strength	Pass?	Dyn.Rng	Legato%	Chroma sim.
0.35	FAIL	19.64	54.8	0.681
0.65	FAIL	18.62	57.7	0.723
0.85	FAIL	14.86	70.8	0.755

Table 2. Higher strength wins on EVERY metric in the tested range — the opposite of the assumed fidelity/compliance trade-off. 0.85 > 0.35 as a starting point.

Real Chroma / MFCC Heatmap — DSP Target vs. AI Output

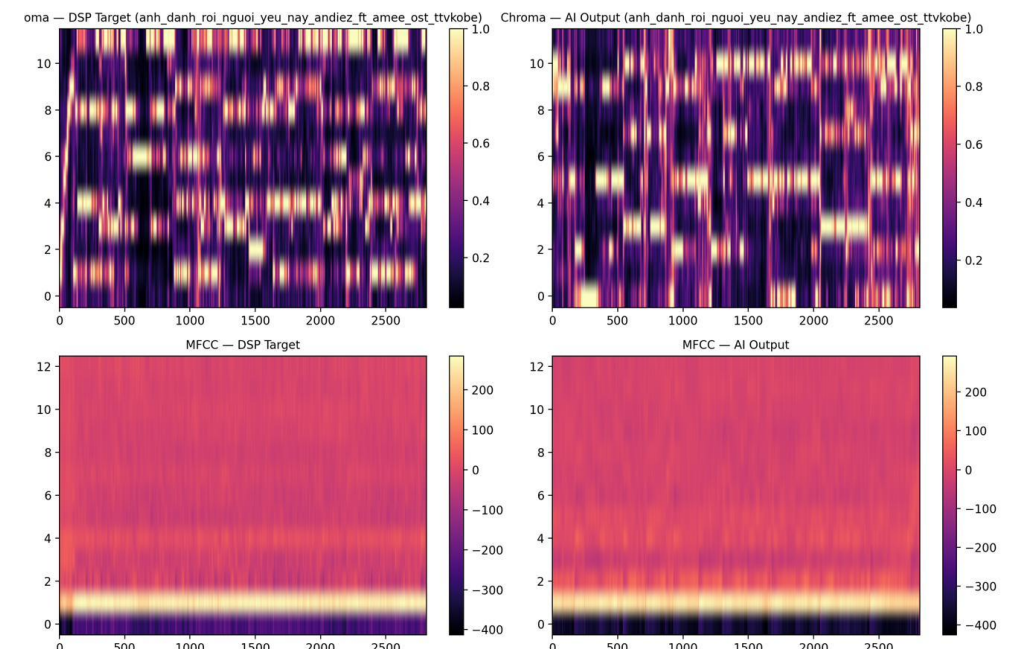


Figure 1c. Visually near-identical chroma (melody/harmony) and MFCC (timbre) patterns — mean similarity 0.84 (chroma) and 0.98 (MFCC) across 10 songs.

Ablation Study — an honest negative finding

--ablate_input_audio zeroes InputAudioProjection's tokens, same seed — isolating source-audio conditioning's contribution.

Normal vs. ablated: cosine similarity 1.0000, chroma/MFCC DTW cost ~0.0001 — nearly IDENTICAL audio. img2img sampling at strength 0.65 likely already anchors structure so strongly that the cross-attention module's added contribution isn't detectable at this strength. Not the confirmation hoped for — reported honestly, with a lower-strength re-test proposed as follow-up.

KNOWN LIMITATIONS

- Legato is inconsistent, not simply low: median often already lands in the 70–80% band (73.6% avg, in-sample), but IQR of 30–49 points means transitions swing wildly — a variance problem, not a ceiling.
- Ablation didn't detect InputAudioProjection's effect at strength 0.65 (see above) — needs a lower-strength re-test before concluding either way.
- Imperfect vocal isolation: Demucs quality varies by track.
- Strength range tested (0.35–0.85) trends toward better compliance at the top end — higher values untested.

5 CONCLUSION & FUTURE WORK

- Evaluated on 10 unseen + 10 in-sample songs: the model generalizes cleanly on 5/7 compliance dimensions regardless of training exposure, and Stage 3 reliably fixes dynamic range on BOTH populations (0-1/10 → 10/10) — direct evidence the deterministic safety-net design works as intended.
- Legato-overlap correction remains inconsistent on both populations, and the ablation study could not detect the source-audio conditioning module's contribution at the tested strength — both reported directly rather than smoothed over.
- Future work: retune legato correction for variance (not just average); re-run the ablation at lower strength; extend the strength sweep beyond 0.85; extended training; CFG negative-prompting against vocals; an agentic front-end for automatic karaoke-source discovery.

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